

Audit analytics and automation: 25% faster data preparation

n8n workflows and a Power BI model to pull and reconcile audit data for forensic and controls engagements.

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Client names and identifying details have been withheld. Prepared by Archish Chugh for portfolio purposes; this is not an EY publication and does not represent the views of any client.

Audit analytics and automation: 25% faster data preparation

Set up n8n and Power BI workflows to pull and reconcile audit data 25% faster. This cut engagement preparation time and gave more time for actual control testing.

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USED ON

EY Risk & Compliance engagements

ROLE

Designed and built the workflows

TOOLS

n8n · Power BI · Excel

RESULT

25% faster data preparation

EXECUTIVE SUMMARY

SITUATION

Client data on forensic and controls engagements came in many formats: PDFs, ERP exports, registers and spreadsheets.

CHALLENGE

Collecting and matching this data by hand slowed every engagement and left less time for testing.

APPROACH

Built n8n workflows to collect and reconcile the data, added exception rules, and set up a Power BI model for review.

RESULT

Audit data pulled and reconciled 25% faster. Used on 2–3 engagements, with full-population testing where the data allows.

25%

faster data pull and reconciliation

2–3

engagements using the workflows so far

100%

of transactions tested, instead of samples, where the data allows

01 · THE PROBLEM

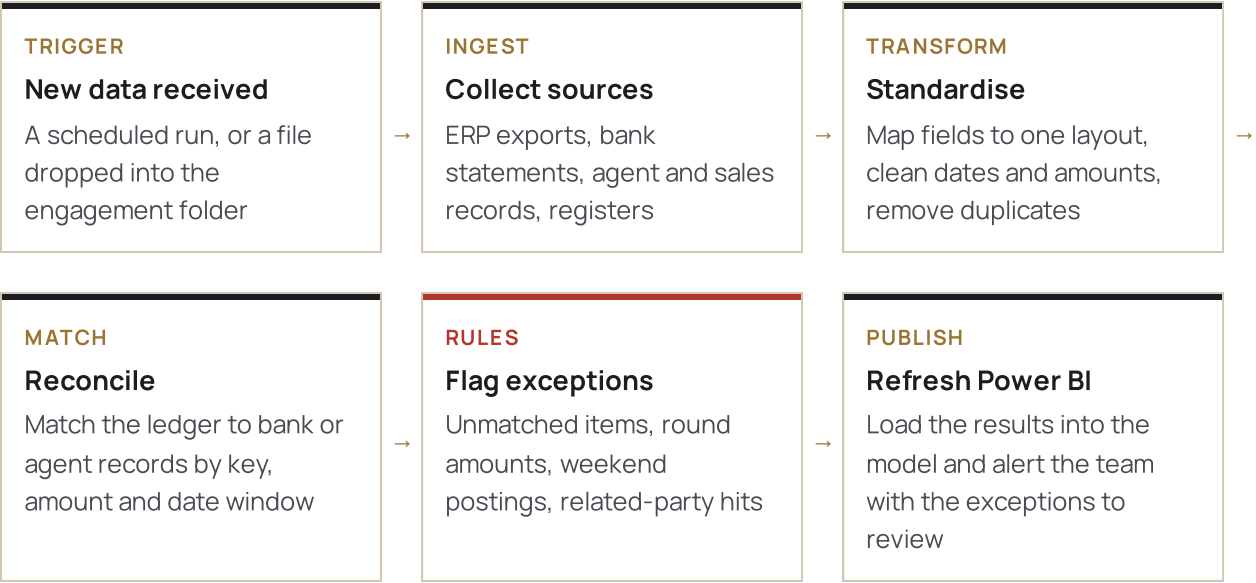
Background

On forensic and controls engagements, data arrives in many forms: bank statements as PDFs, ERP exports, agent sales files, collection registers and emailed spreadsheets. Analysts spent days collecting, reformatting and matching it by hand. This was slow, prone to errors and done differently on each engagement, and it reduced the time left for control testing.

n8n workflow

Exhibit 1: Reconciliation workflow, from source to exception list

Representative design.



Here is how it worked on the aviation engagements. The workflow matched counter ticket sales against agent records and matched cash collections against the books. The exceptions it produced, such as collections never recorded and excess baggage not charged, became the starting point for control testing instead of something the team found at the end.

Working demo

This is a small version of the matching and exception rules from the workflow. It uses sample data, not client records. Turn rules on or off to see how the exception list changes.

Exhibit 2: Collections, books against bank (sample)

Sample data for demonstration

9 items tested 3 matched 6 exceptions

DATE	REFERENCE	BOOKS ₹	BANK ₹	RESULT
Mon, 03 Mar	Agent A collection	48,250	48,250	Matched
Tue, 04 Mar	Agent B collection	1,20,000	1,20,000	Round amount
Wed, 05 Mar	Counter sales	36,780	35,780	Differs by ₹1,000
Sat, 08 Mar	Voucher V-114	25,500	25,500	Weekend posting
Mon, 10 Mar	Agent C collection	62,410	—	Not in bank
Tue, 11 Mar	Agent A collection	51,090	51,090	Matched
Wed, 12 Mar	Refund R-22	50,000	50,000	Round amount
Thu, 13 Mar	Unidentified receipt	—	18,600	Not in books
Fri, 14 Mar	Agent D collection	44,375	44,375	Matched

04 · DATA MODEL

Power BI data model

Exhibit 3: Star schema behind the dashboards

Representative.

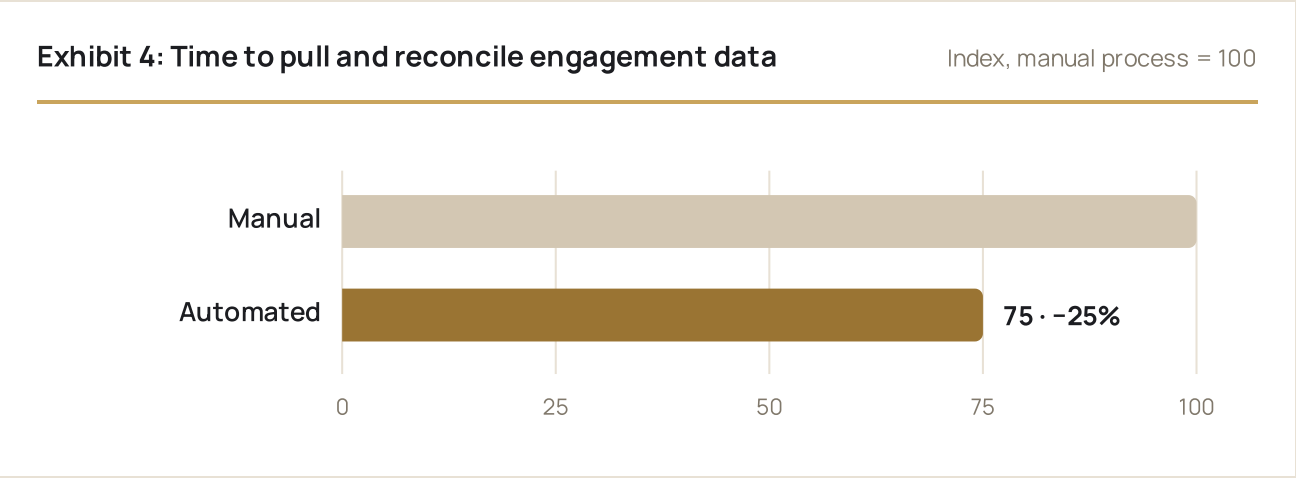
Dim_Entity Legal entities, directors, shared addresses	Fact_Transactions Every ledger and bank line, with a match status and exception flags	Dim_Counterparty Vendors, agents and customers, with related-party tags
Dim_Date Period, fiscal year, weekend and holiday flags	Fact_Exceptions Rule hits, reviewer, disposition and evidence reference	Dim_Control Control ID, owner, COSO component, test result

The dashboards had three views. An engagement overview showed reconciliation coverage and open exceptions. An exception register let reviewers filter by rule, entity and value and record what they decided. A controls view

linked each exception to the control that should have prevented it, so the audit report could go straight from the data to the root cause.

05 · RESULTS

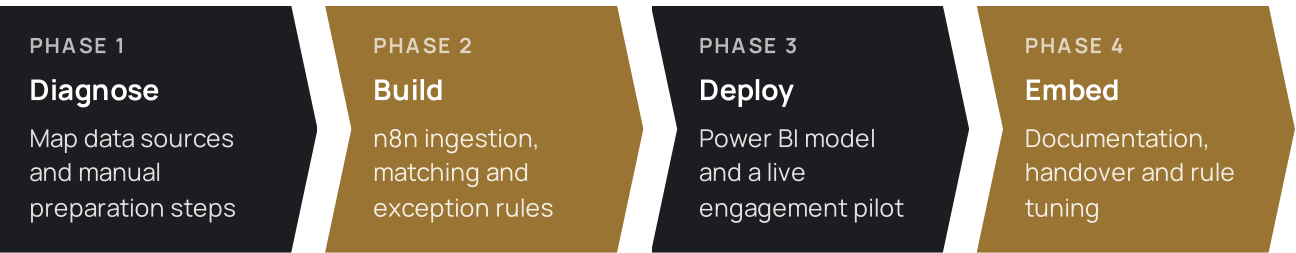
Results



- **Productivity:** preparing engagement data took 25% less time, and that time went into control testing.
- **Assurance:** reconciliations cover full populations, so we test every transaction instead of a sample and exceptions can't fall between samples.
- **Consistency:** the same rules run on every engagement, and each exception links to its evidence reference for review.

06 · ADOPTION

Rollout



- 1 Start with one real engagement**
Built the first workflow on a live engagement, so the time saved was measured on real work.
- 2 Show managers the exceptions**
Walked managers through the exception register and how each flag ties back to a control finding.
- 3 Write it down**
Documented each workflow's inputs, rules and outputs so other analysts can reuse it independently.

4

Improve from feedback

Added and adjusted rules based on reviewers' decisions, which reduced false positives over time.

KEY TAKEAWAYS

- 1 Automating data preparation gives more time for control testing.
- 2 Rules run on the full population catch exceptions that samples can miss.
- 3 A live pilot and clear documentation helped the team start using the workflows.

UAE

Relevance to UAE roles

Internal audit teams in UAE banks, airlines and government entities are adding data analytics and continuous auditing to their work, in line with IIA Global Internal Audit Standards. This experience can be applied with whichever automation or BI tools a team already uses.